

PROFILE Software Engineer with experience in backend development, data engineering, DevOps, and machine learning, spanning across startup, defense contracting, and research environments.

EDUCATION **M.S. Computer Science, May 2025**
Temple University, Philadelphia, PA | 3.83 GPA

B.S. Mechanical Engineering, June 2016
Drexel University, Philadelphia, PA | 3.86 GPA

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript (Node.js), Ruby, SQL, C, Perl, Bash, MATLAB
- **Frameworks:** Flask, Django, FastAPI, SpringBoot, Rails
- **Cloud Development Tools:** AWS, Docker, Kubernetes, Helm, Snowflake
- **Testing Tools and Frameworks:** Selenium, Cucumber, pytest, JUnit, GoogleTest
- **CI / CD:** GitHub, GitLab, Jenkins
- **Database Tools:** psql, mysql, dbt

WORK **Arcadia | Remote**
Late-stage startup utilizing utility data for renewable energy solutions

Senior Software Engineer	April 2024 - Present
Software Engineer III	April 2023 - April 2024
Software Engineer II	April 2022 - April 2023

Growth and Governance Team

- Maintained a utility data pipeline in Java to ensure accurate transformation and normalization of raw utility statement data into PostgreSQL data models
- Implemented a utility meter reconciliation feature to match meters without clear identifiers across utility statements
- Developed data warehouse models from project inception, implementing Dimension, Fact, and Aggregate tables to consolidate data from disparate sources
- Implemented the migration of a legacy data transfer tool to the data warehouse backend, leading the delivery of customer-facing shell scripts for schema and data loading
- Improved scalability by building a Kubernetes memory optimization tool and establishing a CI/CD pipeline via GitHub Actions and Helm

Residential Utility Data Team

- Developed utility connectors in Python that authenticated users and captured residential utility data from consumer utility portals and websites
- Led an effort to move utility connectors from running on AWS Lambda to Kubernetes
- Updated Rails models, controllers, and processors for managing data in a Postgres database
- Increased the scope and size of unit tests, as well as giving several internal presentations to enable other team members to do the same

The Stratagem Group | King of Prussia, PA
Small-scale contractor for various programs spanning across the DoD and Intelligence Community

Software Engineer Senior	August 2019 - April 2022
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App Framework & Infrastructure Contract

- Defined and implemented a program-wide DevSecOps pipeline that could build, test, scan, and package C++, Java, and Python applications through GitLab CI/CD
- Developed libraries in C++, Java, and Python to simplify API calls to a customer SDK
- Fixed library linking issues that occurred when compiling C++ code using Autotools
- Found and resolved bugs by deploying and testing applications in an OpenShift pipeline

Automated Testing Contract

- Created an AWS Lambda Python application to export / import TestRail project data
- Developed automated UI tests using Cucumber.js and Selenium to simulate user actions for over 40 different test scenarios
- Created Python, Java, and Node.js automated test pipeline templates in Jenkins

Cloud Development Contract

- Designed and implemented a backend REST API query service using Spring Boot
- Implemented unit tests using JUnit and Mockito

Machine Learning Contract

- Created and improved functions for reading and classifying incoming data from an Apache Kafka broker
- Developed methods for hyperparameter optimization utilizing Optuna

Co-op Program Recruitment Lead

- Led the process for recruiting and hiring students from the Drexel University Co-op program

Lockheed Martin, Space Systems Company | King of Prussia, PA

Large-scale contractor for various programs spanning across the DoD and Intelligence Community

Software Engineer

April 2018 - August 2019

Associate Software Engineer

September 2016 - April 2018

- Automated software installation and server configuration tasks
- Developed Perl scripts to synchronize software packages on an air-gapped system
- Created and improved End-to-end tests using Selenium and C#

RESEARCH

Capstone Project | Temple University

Approximation Algorithms for the Traveling Salesman Problem

January - May 2025

- Implemented state of the art algorithms for the Traveling Salesman Problem, including Ant Colony Optimization and Q-Learning, and compared efficacy on several problem types

Center for Data Analytics and Biomedical Informatics | Temple University

Machine Learning to Detect Hypertensive Retinopathy

August 2022 - May 2023

- Developed a PyTorch RNN pipeline to classify retinal images for indicators of hypertension
- Improved the model via hyperparameter tuning, oversampling, and image augmentation

Center for Data Analytics and Biomedical Informatics | Temple University

Social Media Bias Identification

June 2022 - August 2022

- Parsed data from articles, Tweets, and Reddit posts related to the Russo-Ukrainian War
- Mined novel data from Reddit related to the Russo-Ukrainian War